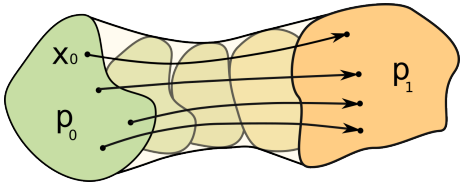


Flow matching (FM)

[simulation-free training of generative ordinary differential equations]

Skoltech
Skolkovo Institute of Science and Technology

AIRI

Outline

The task of generative modeling

Diffusion models and their limitations

Flow matching

Rectified flows:
straightening the trajectories

Flow matching
with general transport plans

Extensions of flow matching

Conclusion

The task of generative modeling

Modern Generative Models for Images

Text prompt: woman's transparent futuristic inspired sneakers, glitter, depth of field

KANDINSKY

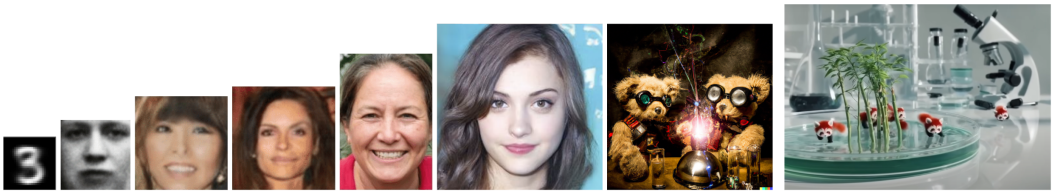
Text prompt: Chicken with potatoes baked in mayonnaise-sour cream sauce

SHEDEVRUM

Text prompt: 1967 Dodge Charger, moody lighting, side view, black, front view, lobby of the Louvre ...

MIDJOURNEY

Evolution of Generative Models



2013-2014
Initial attempts to learn generative models; VAE/GAN algorithms

2015-2017
Active development of (primarily GAN-based) algorithms, first convincingly scalable image generation models appear.

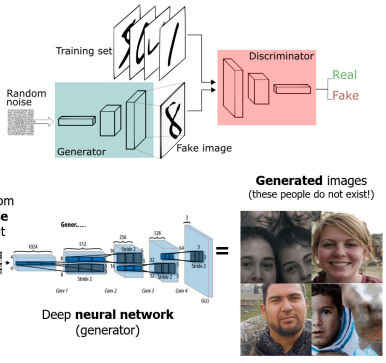
2019-2021
Diffusion-based models are introduced and actively developed.

2022
First inspiring large **text-to-image** models appear (based on diffusion models).

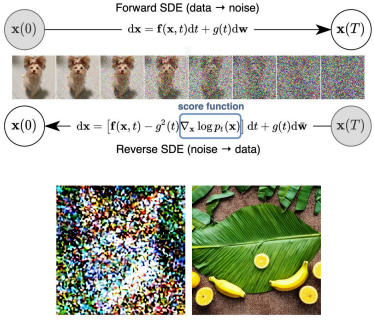
2024
Powerful **text-to-video** models appear based on diffusion models.

Principal Approaches to Generative Modeling[1][2]

Adversarial models (GANs, 2014)



Diffusion Models (DM, 2019)



[1] Ian Goodfellow et al. (2014). **"Generative adversarial nets".** In: *Advances in neural information processing systems*, pp. 2672–2680.

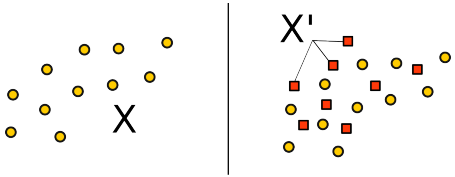
[2] Jascha Sohl-Dickstein et al. (2015). **"Deep unsupervised learning using nonequilibrium thermodynamics".** In: *International conference on machine learning*. PMLR, pp. 2256–2265.

The key task of generative modeling (unconditional generation)

The informal task:[3]

given a data sample X, provide a procedure

to synthesize **new** data X' which is similar to the given data.

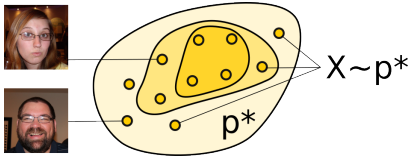


An example of a data sample X

[3] This is the **basic** generative modeling task formulation. In practice, one usually requires not just synthesising new samples but samples depending on some **additional condition**, e.g., text (text-to-image, etc.).

The unconditional data generation

The key assumption: the data samples $X = \{x_1, \dots, x_N\}$ are random i.i.d. samples from some *unknown* data distribution p^* (accessible only via the empirical samples X), e.g., on $\mathbb{R}^D$.



The formal task: estimate the data distribution p^* given an empirical sample $X \sim p^*(\cdot)$. More precisely, one want to be able to do data generation (sampling), i.e., one has to

provide a procedure to sample $x^{new} \sim p^*(\cdot)$.

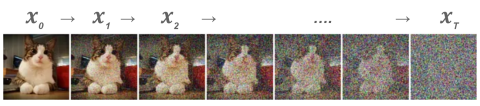
Diffusion models and their limitations

Diffusion Models[4][5]

MAIN IDEA: reverse the data noising process.

Forward diffusion (noising SDE)

Take a data distribution $x_0 \sim p_0$ $(= p^*)$ and gradually turn it to noise distribution $x_T \sim p_T = \mathcal{N}(0, \sigma^2 I)$.



$$dx_t = f(x_t, t)dt + g(t)dW_t$$

(e.g., $dx_t = -\frac{1}{2}\beta_t dt + \sqrt{\beta_t} dW_t$)

Reverse diffusion (denoising SDE)

Sample from noise distribution $x_T \sim p_T$ and reverse the diffusion to get $x_0 \sim p_0$:

$$dx_t = [f(x_t, t) - g^2(t)\nabla_x \log p(x_t, t)]dt + g(t)d\overline{W_t}$$

(or $dx_t = [f(x_t, t) - \frac{1}{2}g^2(t)\nabla_x \log p(x_t, t)]dt$)

[4] Jonathan Ho, Ajay Jain, and Pieter Abbeel (2020). **"Denoising diffusion probabilistic models".** In: *Advances in neural information processing systems* 33, pp. 6840–6851.

[5] Yang Song et al. (2020). **"Score-Based Generative Modeling through Stochastic Differential Equations".** In: *International Conference on Learning Representations*.

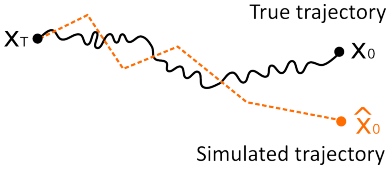
The Key Limitation of Diffusion Models: Time-Consuming Inference

To simulate the denoising process:

$$\mathrm{d}x_t = \left[f(x_t, t) - g^2(t)\nabla_x \log p(x_t, t)\right]\mathrm{d}t + g(t)\mathrm{d}\overline{W}_t$$

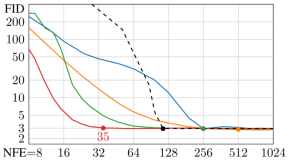
one uses the discretization (e.g., Euler-Maruyama simulation):

$$x_{t-\Delta t} = x_t - \left[f(x_t, t) - g^2(t)\nabla_x \log p(x_t, t)\right]\Delta t + g(t)\sqrt{\Delta t}\xi_t \quad , \xi_t \sim \mathcal{N}(0, I).$$



Remark:

NFE (# function evaluations) ≡ (# discretization steps)



Diff. models performance, CIFAR-10. FID w.r.t NFE.

Straightening The Trajectories of Diffusion Models

What we have

Not straight (deterministic or stochastic) trajectories, which are HARD to simulate.

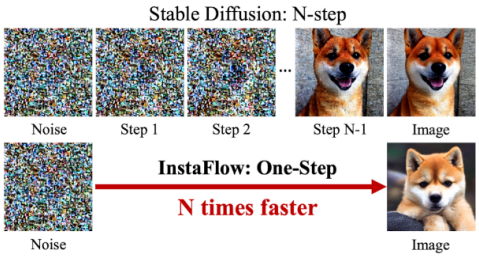
What we want

Straight (deterministic?) trajectories, which are EASY to simulate.

Flow matching

Teaser: Flow Matching Capabilities[6][7]

Insta**Flow**: 1 Step is Enough for HQ Diffusion-based Text-to-image Synthesis



Scaling Rectified **Flow** Transformers for High-Resolution Image Synthesis

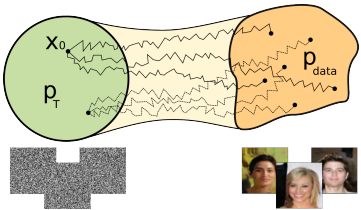
[6]Xingchao Liu, Xiwen Zhang, et al. (2023). **"Instaflow: One step is enough for high-quality diffusion-based text-to-image generation".** In: *The Twelfth International Conference on Learning Representations*.

[7]Patrick Esser et al. (2024). **"Scaling rectified flow transformers for high-resolution image synthesis".** In: *Forty-first International Conference on Machine Learning*.

Flow matching vs. Diffusion Models: Key Differences

Diffusion models framework (2019)

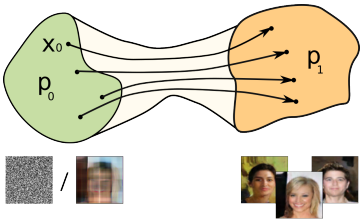
- maps given complex data distribution to the normal distribution.



- uses pre-defined noising process.
- (theoretically) requires infinite time horizon $[0, T]$ (in some cases).
- based on SDEs ($\Rightarrow$ complex stuff).

Flow matching framework (2023)

- maps arbitrary distribution p_0 to arbitrary distribution p_1.



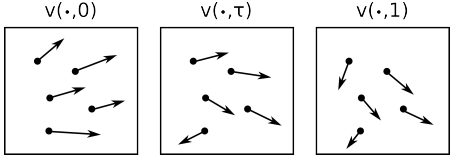
- no pre-defined process.
- finite time horizon $[0, 1] = [0, T]$.
- based on ODEs.

Main flow-matching related papers

1. **Flow Matching (FM):** Yaron Lipman et al. (2022). **"Flow Matching for Generative Modeling".** In: *The Eleventh International Conference on Learning Representations*
2. **Rectified Flows (RF):** Xingchao Liu, Chengyue Gong, et al. (2022). **"Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow".** In: *The Eleventh International Conference on Learning Representations*
3. **Conditional FM (OT-CFM):** Alexander Tong et al. (2024). **"Improving and generalizing flow-based generative models with minibatch optimal transport".** In: *Transactions on Machine Learning Research*. Expert Certification. ISSN: 2835-8856. URL: https://openreview.net/forum?id=CD9Snc73AW
4. **Straightening FM:** Aram-Alexandre Pooladian et al. (2023). **"Multisample Flow Matching: Straightening Flows with Minibatch Couplings".** In: *International Conference on Machine Learning*. PMLR, pp. 28100–28127
5. **Optimal FM (OFM):** Nikita Kornilov et al. (2024). **"Optimal Flow Matching: Learning Straight Trajectories in Just One Step".** In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. URL: https://openreview.net/forum?id=kqmucDKVcU

Reminder: vector fields and movement along it

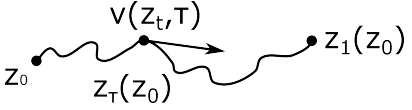
Vector field $v : \mathbb{R}^D \times [0,1] \to \mathbb{R}^D$.



Movement of a point along the field.

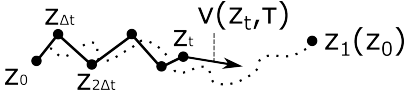
Let $z_t(z_0)$ be the solution to $\mathrm{d}z_t = v(z_t, t)\mathrm{d}t$ with initial condition $z_{t=0} = z_0$, i.e.:

$$z_t(z_0) = z_0 + \int_0^t v(z_\tau(z_0), \tau)\mathrm{d}\tau.$$

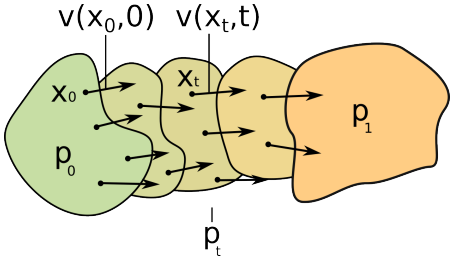


Practical simulation of the movement (Euler solver).

$$z_{t+\Delta t} = z_t + v(z_t, t)\Delta t.$$

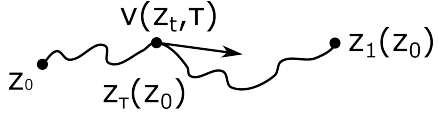


Flow transport: key idea



Just like in the original CNF, we want to find a (time-dependent) vector field $v(x_t, t)$ which transports the probability mass of distribution p_0 to distribution p_1, i.e.:

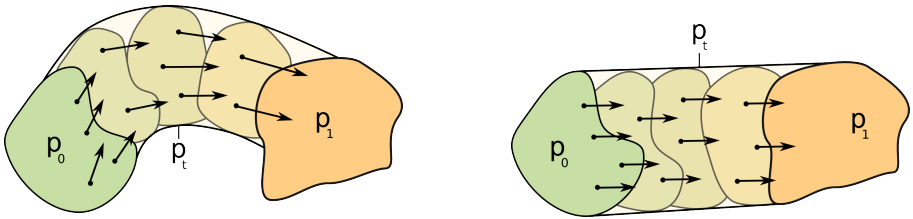
If $x_0 \sim p_0(\cdot)$ then $x_1(x_0) \sim p_1(\cdot)$



We allow general input distributions p_0, not necessarily with the tractable density. Also, from now on we switch to a more general notation (x_0, x_1) rather than (z, x) as before.

Flow transport: non-uniqueness

There may exist multiple vector fields v transporting p_0 to p_1.



Let p_t denote the distribution of $x_t(x_0)$ (for $x_0 \sim p_0(\cdot)$) obtained from p_0 by transporting its mass along the vector field $v(x_t, t)$.

How to construct at least one sequence of distributions p_t which transports p_0 to p_1?

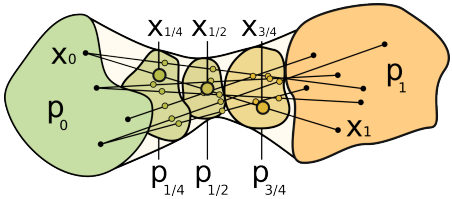
Remark: This should be much easier than constructing vector field v itself.

Flow transport: creating interpolating curve

Simple interpolation. Pick $x_0 \sim p_0(\cdot)$, $x_1 \sim p_1(\cdot)$ and set:

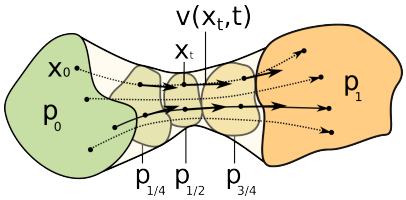
$$x_t = x_0 \cdot (1 - t) + x_1 \cdot t$$

Let p_t be the distribution of points x_t obtained with the procedure above.



How to find some vector field $v(x_t, t)$ which produces this sequence of distributions p_t by transporting p_0?

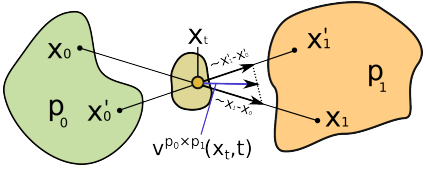
Flow matching: approach algorithm[8]



$$\begin{cases} \mathrm{d}x_t = v(x_t, t)\mathrm{d}t \\ x_0 \sim p_0(\cdot) \end{cases}$$

One of such vector fields could be found as a solution to flow matching (FM) objective:

$$\min_{v} \mathop{\mathbb{E}}_{\substack{x_0 \sim p_0 \\ x_1 \sim p_1}} \left\{ \mathop{\mathbb{E}}_{t \sim [0,1]} \Big\| v(x_t, t) - \underbrace{(x_1 - x_0)}_{x_0 \cdot (1-t) + x_1 \cdot t} \Big\|^2 \right\}.$$



Let $v^{p_0 \times p_1}(x_t, t)$ denote the minimizer to this problem.

[8] Xingchao Liu, Chengyue Gong, et al. (2022). **"Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow".** In: *The Eleventh International Conference on Learning Representations*.

The marginal-preserving field of flow matching[9]

Theorem (Marginal preserving vector field in FM)

For $t \in [0,1]$ define $p_t = \mathrm{Law}(x_t)$ to be the distribution of the random variable $x_t = (1-t)x_0 + tx_1$, where $x_0 \sim p_0(\cdot)$ and $x_1 \sim p_1(\cdot)$ are independent. Let $p(x_0, x_t, x_1)$ denote the joint distribution of x_0, x_t, x_1. Define the vector field

$$v^{p_0 \times p_1}(x_t, t) = \mathbb{E}_{(x_0,x_1)\sim p(x_0,x_1|x_t)}[x_1 - x_0] = \int [x_1 - x_0] p(x_0, x_1|x_t) dx_0 dx_1$$

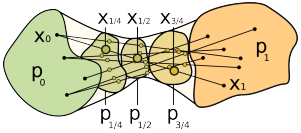
Let $\{x'_t(x_0)\}_{t=0}^1$ be the solution of ODE $dx'_t = v^{p_0 \times p_1}(x'_t, t)dt$ with initial condition $x'_0(x_0) = x_0$. Then for all $t \in [0,1]$ it holds that

$$\mathrm{Law}(x'_t(x_0)) = \mathrm{Law}(x_t) = p_t,$$

*i.e., **movement along the field $v^{p_0 \times p_1}$ maps p_0 to p_1 and preserves all the time-marginal distributions p_t in the interpolating sequence $\{p_t\}_{t=0}^1$.***

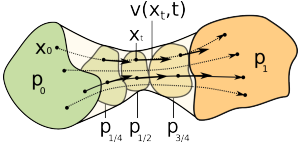
Before FM

(no vector field)



After FM

(constructed vector field)



[9] Xingchao Liu, Chengyue Gong, et al. (2022). **"Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow".** In: *The Eleventh International Conference on Learning Representations*.

How to get the FM's marginal preserving vector field?

The vector field of FM is given by the following expression:

$$v^{p_0 \times p_1}(x_t, t) = \mathbb{E}_{(x_0, x_1) \sim p(x_0, x_1 | x_t)}[x_1 - x_0] = \int [x_1 - x_0] p(x_0, x_1 | x_t) dx_0 dx_1.$$

Question

If we know an expression for $v^{p_0 \times p_1}(x_t, t)$, then why do we need to solve an optimization problem (recall previous slides) to get $v^{p_0 \times p_1}$?

Unfortunately, one may not easily compute or estimate the expression above as it requires sampling from $x_0, x_1 | x_t$, which is intractable. Indeed, we can only sample $x_t | x_0, x_1$.

Proposition (Tractable objective for recovering the vector field)

$$v^{p_0 \times p_1} = \arg\min_{v} \mathbb{E}_{\substack{x_0 \sim p_0 \\ x_1 \sim p_1}} \left\{ \mathbb{E}_{t \sim [0,1]} \left\| v(x_t, t) - (x_1 - x_0) \right\|^2 \right\}$$

where $x_t = x_0 \cdot (1 - t) + x_1 \cdot t$

Flow matching: practical aspects[10]

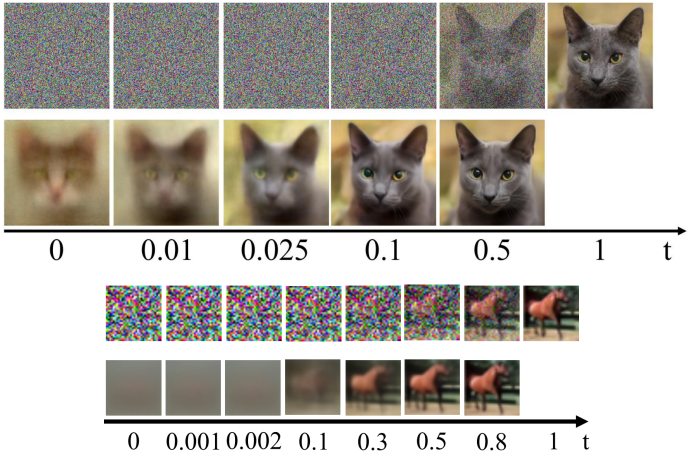
$$v_{\theta^*} = \arg\min_{v_\theta} \mathop{\mathbb{E}}_{\substack{x_0 \sim p_0 \\ x_1 \sim p_1}} \left\{ \mathop{\mathbb{E}}_{t \sim [0,1]} \left\| v_\theta(x_t, t) - \underbrace{(x_1 - x_0)}_{x_0 \cdot (1-t) + x_1 \cdot t} \right\|^2 \right\}$$

- $\mathbb{E}_{x_0 \sim p_0(\cdot), x_1 \sim p_1(\cdot)} \left[\cdot \right]$ is estimated using train samples $x_0 \sim p_0$ and $x_1 \sim p_1$ (datasets).
- Time t is sampled at random from Uniform$[0, 1]$.
- Vector field $v = v_\theta$ is a neural network which takes two inputs: data point x_t and time (time embedding) t as the input.

[10] Yaron Lipman et al. (2022). **"Flow Matching for Generative Modeling".** In: *The Eleventh International Conference on Learning Representations*.

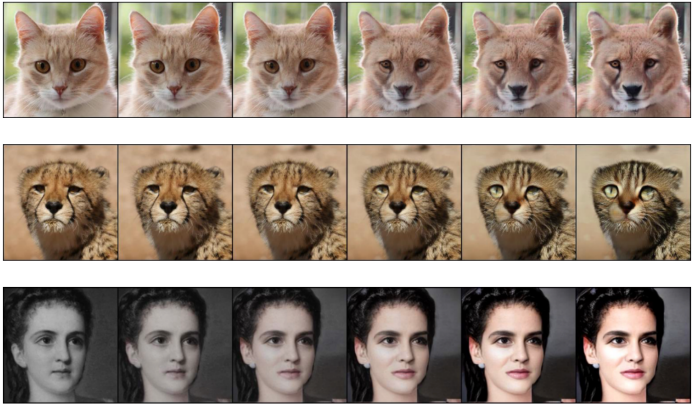
Flow matching: noise-to-data generation & trajectories

- The 1st line in each plot shows the learned trajectory satisfying $dx_t = v_\theta(x_t, t)dt$.
- The 2nd line shows naive endpoint prediction $x_t \mapsto x_t + v_\theta(x_t, t)(1-t)$ from x_t.



Flow matching: unpaired data-to-data translation

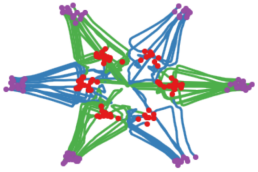
Trajectories of **unpaired domain translation** (between various domains) learned by FM.



Note: this task is feasible for FM but, in general, infeasible for CNF with log-likelihood training as the input distribution p_0 does not have readily available density $p_0(x_0)$.

Flow matching: straightness issues

Toy example:
Gaussians -> Gaussians



Example of image generation:
(different number of trajectory integration steps is shown)

Remark: if the trajectories were really straight, generation with different numbers of steps would give the same result, because there would be no integration errors.

Problem: trajectories are not straight enough.

Question

Can we make the flow trajectories **straighter**? In practice, straight trajectories require less numerical integration steps, e.g., in the Euler scheme, so they are desirable.

Rectified flows:
straightening the trajectories

Rectified flows[11]: key idea

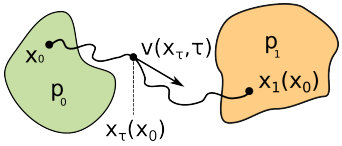
Key idea: iteratively repeat flow matching (FM) procedure to get straighter trajectories.

Step 1: pure flow matching.

$$v^1 = \arg\min_v \mathbb{E}_{\substack{x_0 \sim p_0 \\ x_1 \sim p_1}} \left\{ \mathbb{E}_{t \sim [0,1]} \left\| v(x_t, t) - \underbrace{(x_1 - x_0)}_{\| \; x_0 \cdot (1-t) + x_1 \cdot t} \right\|^2 \right\}.$$

Result: vector field v^1 moving p_0 to p_1:

$$x_0 \sim p_0 \wedge \mathrm{d}x_t = v^1(x_t, t)\mathrm{d}t \quad \Longrightarrow \quad x_1^{v^1}(x_0) \sim p_1.$$



[11] Xingchao Liu, Chengyue Gong, et al. (2022). **"Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow".** In: *The Eleventh International Conference on Learning Representations*.

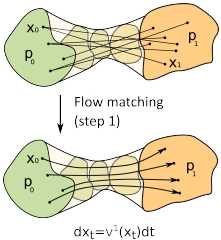
Rectified flows

Step k+1: flow matching using samples $(x_0, x_1^{v^k}(x_0))$:

$$v^{k+1} = \arg\min_{v} \mathbb{E}_{x_0 \sim p_0} \left\{ \mathbb{E}_{t\sim[0,1]} \left\| v(x_t, t) - (x_1^{v^k}(x_0) - x_0) \right\|^2 \right\}.$$

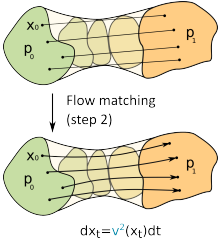
$x_0 \cdot (1-t) + x_1^{v^k}(x_0) \cdot t$

Step 1



Learn some vector field v^1 transporting p_0 to p_1.

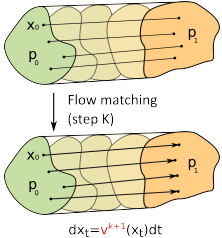
Step 2



Use the previously obtained field v^1 to get samples $(x_0, x_1^{v^1}(x_0))$ to train the new field v^2.

...

Step k+1

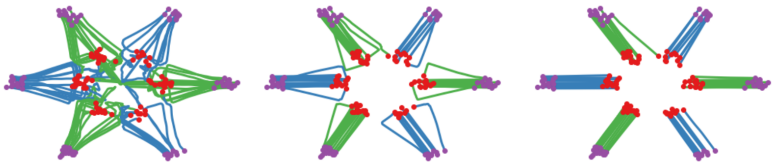


Use the previously obtained field v^k to get samples $(x_0, x_1^{v^k}(x_0))$ to train the new field v^{k+1}.

Rectified flows: the straightening effect

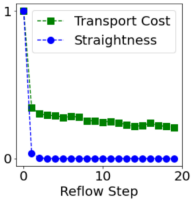
Theorem (*Informal*[12])

Vector field v^K *produces more straight trajectories* ($\mathrm{d}x_t = v^K(x_t, t)\mathrm{d}t \wedge x_0 \sim p_0$) *as* $K \to \infty$.



(a)The 1st rectified flow $\boldsymbol{Z}^1$ (b) Reflow $\boldsymbol{Z}^2$ (c) Reflow $\boldsymbol{Z}^3$

Rectified Flows performance between source and target, $K = 1, 2, 3$.

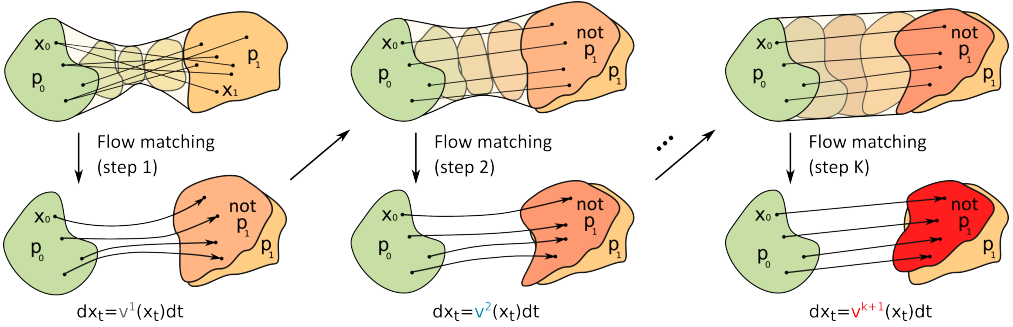


Transport cost, Straightness.

[12]Xingchao Liu, Chengyue Gong, et al. (2022). **"Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow".** In: *The Eleventh International Conference on Learning Representations*.

Rectified flows: practical issues

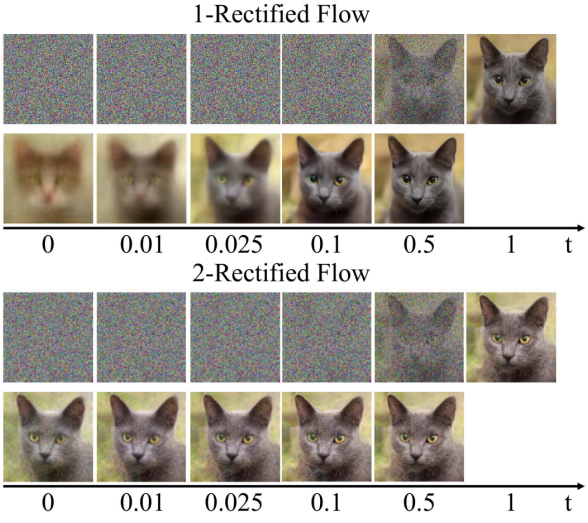
After the first flow matching step, the iterative approach **never** utilizes the true data $x_1 \sim p_1(\cdot)$ but uses learned samples $x_1^{v^k} \approx p_1(\cdot)$, i.e., those obtained via $x_0 \sim p_0(\cdot) \wedge \mathrm{d}x_t = v^k(x_t, t)\mathrm{d}t$.



This issue leads to target matching error, where the model never learns the true p_1 due to accumulating statistical, approximation, and optimization errors across FM iterations.

Furthermore, each iteration after the first one requires simulation an ODE from the previous step $x_0 \sim p_0 \wedge \mathrm{d}x_t = v^k(x_t, t)\mathrm{d}t$. This makes training not simulation free.

Rectified flows: trajectories of image generation

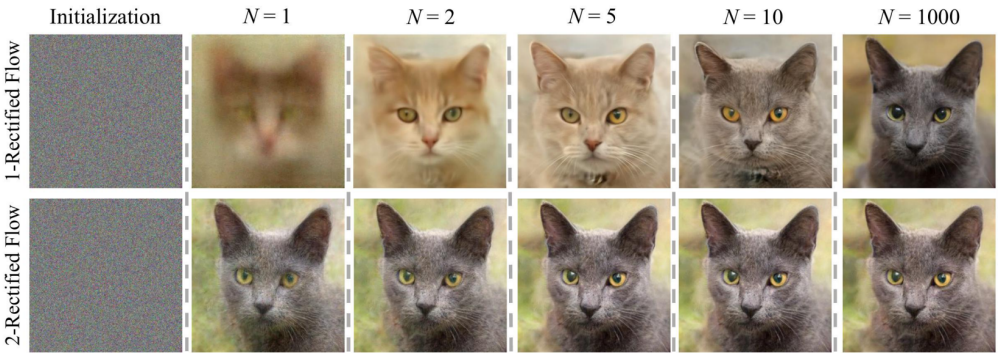


- The 1st line in each plot shows the learned trajectory satisfying $dx_t = v_\theta(x_t, t)dt$.
- The 2nd line shows naive endpoint prediction from x_t: $x_t \mapsto x_t + v_\theta(x_t, t)(1 - t)$.

The trajectory after the second iteration of FM is rather straight. This can be judged by the fact that the last line contains almost the same images.

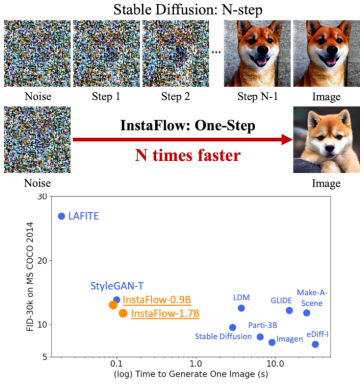
Rectified flows: image generation examples

Columns - different number of integration steps, **rows** – different FM iterations.



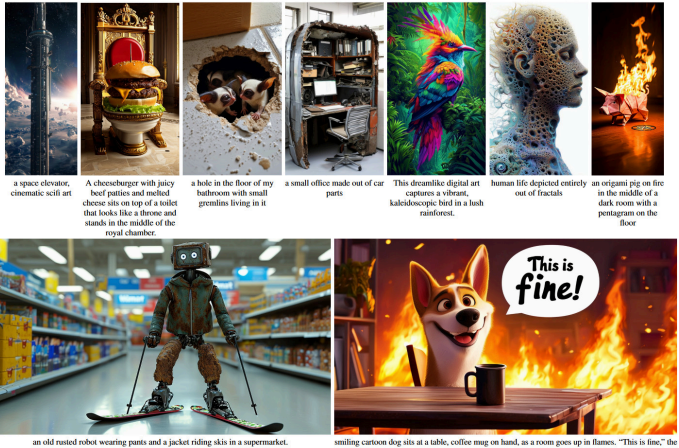
For the 2-iteration FM the trajectories are rather straight which can be deduced from the fact that the model works reasonable with just 1 Euler trajectory integration step (see 2-Rectified flow, $N = 1$).

Instaflow: Rectified Flow as a Tool for Model Distillation[13]



[13]Xingchao Liu, Xiwen Zhang, et al. (2023). **"Instaflow: One step is enough for high-quality diffusion-based text-to-image generation".** In: *The Twelfth International Conference on Learning Representations*.

Stable Diffusion: Flow-based Large Text-to-Image Model[14]



[14] Patrick Esser et al. (2024). **"Scaling rectified flow transformers for high-resolution image synthesis".** In: *Forty-first International Conference on Machine Learning*.

Flow matching
with general transport plans

Extensions of flow matching?

1. **(Initial) Flow Matching:**

$$\min_{v} \mathop{\mathbb{E}}_{\substack{x_0 \sim p_0 \\ x_1 \sim p_1}} \left\{ \mathop{\mathbb{E}}_{t \sim [0,1]} \left\| v(x_t, t) - (x_1 - x_0) \right\|^2 \right\}$$

$$x_t = x_0 \cdot (1-t) + x_1 \cdot t$$

2. **(Iterative) Flow Matching:** Uses dependent (x_0, x_1) pairs where $x_1 = x_1^{v_k}(x_0)$, $x_0 \sim p_0$:

$$\min_{v} \mathop{\mathbb{E}}_{x_0 \sim p_0} \left\{ \mathop{\mathbb{E}}_{t \sim [0,1]} \left\| v(x_t, t) - (x_1^{v_k}(x_0) - x_0) \right\|^2 \right\}$$

$$x_t = x_0 \cdot (1-t) + x_1 \cdot t$$

Recall that the marginal distribution of $x_1^{v_k}(x_0)$ is p_1 by construction of v^k.

Observation

In both cases we sample pairs (x_0, x_1) from some transport plans between p_0 and p_1.

Question

Can we use arbitrary transport plans $\pi \in \Pi(p_0, p_1)$ between p_0 and p_1?

The marginal-preserving field of flow matching revisited

Theorem (Marginal preserving vector field in FM revisited)

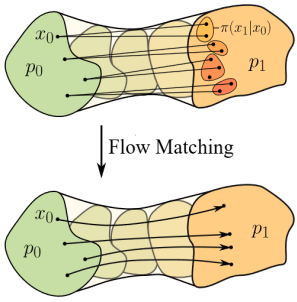
For $t \in [0,1]$ define $p_t = \mathrm{Law}(x_t)$ to be the distribution of the random variable $x_t = (1-t)x_0 + tx_1$, where $(x_0, x_1) \sim \pi$ with $\pi \in \Pi(p_0, p_1)$. Let $p(x_0, x_t, x_1)$ denote the joint distribution of x_0, x_t, x_1. Define the vector field

$$v^{\pi}(x_t, t) = \mathbb{E}_{(x_0,x_1)\sim p(x_0,x_1|x_t)}[x_1 - x_0] = \int [x_1 - x_0] p(x_0, x_1|x_t) dx_0 dx_1$$

Let $\{x'_t(x_0)\}_{t=0}^1$ be the solution of ODE $dx'_t = v^{\pi}(x'_t, t)dt$ with initial condition $x'_0(x_0) = x_0$. Then for all $t \in [0,1]$ it holds that

$$\mathrm{Law}(x'_t(x_0)) = \mathrm{Law}(x_t) = p_t,$$

*i.e., **movement along the field v^{π} maps p_0 to p_1 and preserves all the time-marginal distributions p_t in the interpolating sequence $\{p_t\}_{t=0}^1$.***



Note that the interpolating sequence $\{p_t\}_{t=0}^1$ is, in fact, depends on the chosen transport plan $\pi \in \Pi(p_0, p_1)$. The properties of the resulting flow, e.g., straightness, also depend on it.

Flow matching with general transport plans

$$v_{\theta^*} = \arg\min_{v_\theta} \mathbb{E}_{(x_0,x_1)\sim\pi} \left\{ \mathbb{E}_{t\sim[0,1]} \left\| v_\theta(x_t, t) - \underbrace{(x_1 - x_0)}_{x_0\cdot(1-t)+x_1\cdot t} \right\|^2 \right\}$$

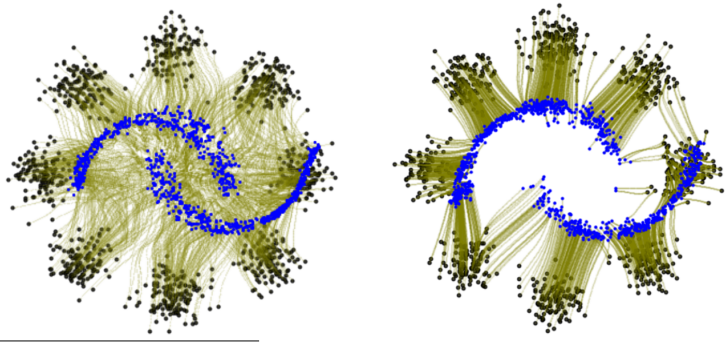
In practice, one typically uses the following transport plans:

1. **Independent Plan:** $x_0 \sim p_0(\cdot), \quad x_1 \sim p_1(\cdot)$ (independent).
   Simple, always available, but may lead to curved trajectories. Still widely used in practice.
2. **Data-Based Plan:** $(x_0, x_1) \sim \pi^*$, where π^* is given (e.g., paired dataset).
   Note, however, that flow matching generally does not preserve the input plan.
3. **Minibatch Optimal Transport (OT):** sample $\{x_0^1, \dots, x_0^N\} \sim p_0$ and $\{x_1^1, \dots, x_1^N\} \sim p_1$ independently, then apply discrete OT solver to optimally pair $\{(x_0^n, x_1^{\sigma(n)})\}_{n=1}^N$.
   Cheap but useful trick. Typically produces a little bit straighter trajectories.

Illustrative examples[15]

Good (bad) transport plan may lead to straight (curved) trajectories of the learned flow.

In the example above, 2 transport plans are compared: **(1)** independent, **(2)** minibatch OT.



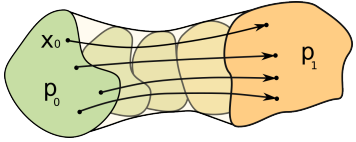
[15]Alexander Tong et al. (2024). **"Improving and generalizing flow-based generative models with minibatch optimal transport".** In: *Transactions on Machine Learning Research*. Expert Certification. ISSN: 2835-8856. URL: https://openreview.net/forum?id=CD9Snc73AW.

Extensions of flow matching

Flow Matching vs. Bridge Matching: a Comparison

Flow Matching

$$\mathrm{d}x_t = v(x_t, t)\mathrm{d}t$$



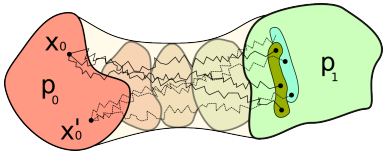
Define interpolation: $x_t \stackrel{\text{def}}{=} x_0 \cdot (1-t) + x_1 \cdot t.$

$$\min_{v} \mathbb{E}_{\substack{x_0 \sim p_0 \\ x_1 \sim p_1}} \mathbb{E}_{t \sim [0,1]} \Big\| v(x_t, t) - (x_1 - x_0) \Big\|^2.$$

Can be iterated to straighten the flows.
Related to the **Optimal Transport** (OT).

Bridge Matching

$$\mathrm{d}x_t = v(x_t, t)\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}W_t \quad (\epsilon > 0).$$



Define a distribution: $p_t^\epsilon \stackrel{\text{def}}{=} \mathcal{N}\big(x_t, \epsilon t(1-t)\big)$

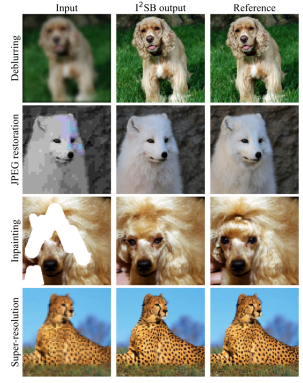
$$\min_{v} \mathbb{E}_{\substack{x_0 \sim p_0 \\ x_1 \sim p_1}} \mathbb{E}_{t \sim [0,1]} \mathbb{E}_{\tilde{x}_t \sim p_t^\epsilon} \Big\| v(\tilde{x}_t, t) - \frac{x_1 - \tilde{x}_t}{1-t} \Big\|^2.$$

Can be iterated and converges to the
Schrödinger bridge.

Examples of Bridge Matching Models for Images[16][17]

Image-to-image Schrodinger Bridge

for various image restoration problems



Diffusion Schrodinger Bridge Matching

for unpaired image-to-image translation

[16]Guan-Horng Liu et al. (2023). **"I2SB: image-to-image Schrödinger bridge".** In: *Proceedings of the 40th International Conference on Machine Learning*, pp. 22042–22062.

[17]Yuyang Shi et al. (2024). **"Diffusion Schrödinger bridge matching".** In: *Advances in Neural Information Processing Systems* 36.

Conclusion

Conclusion

Training made easy

FM lets you train ODEs *without costly trajectory simulations* – pick your pairs and go!

Straighter paths (with a catch)

You can iteratively refine trajectories ($\Rightarrow$ easier inference), but each extra step:

⚠ Requires simulations ⚠ Adds extra errors

Flexible data pairing

Works great whether your $(x_0, x_1) \sim \pi$ pairs are:

✔ Independent ✔ Paired datasets ✔ OT-coupled minibatches

Multiple extensions

Variants like *Bridge Matching* and *Stochastic interpolants*[18] offer even more options.

The trade-off

Fast training... but *slow generation* (thanks to those ODE integrations).

[18] Michael Samuel Albergo and Eric Vanden-Eijnden (2023). **"Building Normalizing Flows with Stochastic Interpolants".** In: *The Eleventh International Conference on Learning Representations*.

Useful Resources

Tutorials

- **Flow Matching Guide and Code**: Comprehensive review of Flow Matching (FM) theory, design choices, and PyTorch examples for image/text generation.
- **Introduction to Flow Matching (Cambridge MLG Blog)**: Beginner-friendly explanation of FM's mathematical foundations and comparisons to CNFs/diffusion models.
- **MIT Course: Flow Matching and Diffusion Models**: Lecture notes, labs, and code for building FM/diffusion models from scratch.

Code implementations

- **Flow Matching Library**: PyTorch library for continuous/discrete FM, including text/image generation examples.
- **TorchCFM: Conditional Flow Matching**: Modular library for OT-CFM, Schrödinger bridges, and image/tabular data generation.